

Practical exam Experiment List for AI

Course Code: **CSL604**

Course Name: **Artificial Intelligence Lab**

Year/Semester: TE / VI

Academic Session: JAN-APRIL 2026

1) Implement DFS search algorithm in Python/Java.
2) Implement DLS search algorithm in Python/Java.
3) Implement DFID search algorithm in Python/Java.
4) Implement Breadth First Search algorithm in Python/Java.
5) Implement Greedy Best First Search algorithm in Python/Java.
6) Implement A* search algorithm in Python/Java.
7) Implement Genetics algorithm in Python/Java. Consider the optimization function $F(x) = ((a+2b+3c+4d)-30)$
8) Implement Hill climbing algorithm in Python/Java. Consider the following equation $y = \sin x$ (or) $-x^2$ (or) $-5x^2+3x+6$
9) Implement crossover operation in Genetics algorithm in Python/Java. Consider a population of 6 chromosomes and convert chromosomes into binary number and perform cross over operation, consider crossover rate as 25 %
10) Implement Mutation operation in Genetics algorithm in Python/Java. Consider a population of 6 chromosomes and convert chromosomes into binary number and perform mutation operation, consider mutation rate as 10 %.
11) Implementing Family tree using prolog.

Note:

- 1) Viva can be asked on entire syllabus Or technical paper that you have taken during practical's.
- 2) Prepare Block world problem, Wumpus world problem and First Order Logic, you can be given any sentence to be converted to FOL
- 3) For search algorithms draw a tree / graph on answer book and write output with open list and close list.
- 4) For Genetic algorithm take 6 chromosomes in population for performing all the operations or specified operation, in output u r expected to print initial population (6 chromosomes)and final population (6 chromosomes) and best chromosome from final population which is optimal solution for equation considered.
- 5) For 9th and 10th experiment perform one iteration.

Subject In charge
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